

Baoren Liu

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Education

University of Illinois Urbana-Champaign, IL
B.S. in Mathematics and Computer Science

Expected May 2028
GPA 3.92/4.00

Experience

Research Intern

June 2024 - Present

Mass General Brigham [medRxiv.org preprint](https://medrxiv.org/preprint)
Boston, MA

- Processed **115+ million** proprietary electronic health records within a **HIPAA-compliant** environment, embedding and migrating clinical notes to vector databases for retrieval augmented generation (**RAG**) pipeline to identify patients with dementia using Python, Pandas, Numpy, Microsoft Azure, **ChromaDB**, **PostgreSQL (pgvector)**, **LangChain**, and **PyTorch**
- Achieved **0.933 F1-score** by experimenting with various RAG techniques, embedding models and LLMs: max marginal relevance retrieval, prompt engineering, BioClinicalBERT, GPT-OSS, Gemma 3, DeepSeek-R1 using **Ollama** and **HuggingFace**
- Benchmarked **RAG** pipelines against traditional ML models (**XGBoost**, **SVM**, **RF**, **LR**) using **TF-IDF** and feature engineering, validating performance via cross-validation and bootstrapping with **Matplotlib** and **Seaborn** visualizations
- Co-authored the research paper evaluating these methodologies, currently under peer-review in Journal of Biomedical Informatics
- Engineered an automated **GitLab CI/CD** pipeline to build and deploy a **Vite/React** [research website](#) via **Plesk** webhook and **SSH**-managed **Linux** runners, incorporating AI-assisted development with **Claude**, **Gemini**, and **Antigravity** agents

Founder & Project Leader

August 2020 - May 2024

Lexington Youth STEAM Team youthsteaminitiative.org
Lexington, MA

- Partnered with local municipal committees to organize Patriots' Day and deploy virtual event platform (**WordPress**, **HTML/CSS/JS**) during COVID-19, processing **200+** community media submissions
- Conducted historical grant trend analysis for the Community Endowment of Lexington using **Python**, **Pandas**, and **NumPy**, identifying key community needs to facilitate a record **\$57,850** funding cycle
- Directed volunteer recruitment by interviewing and onboarding new members, while continuously maintaining the organization's website hosted on **GitHub Pages** across **400+** hours of dedicated community service

Projects

RideList ridelist.app

January 2026 - Present

- Architected a bilingual (English/Chinese) progressive web app (PWA) for campus ridesharing using **Next.js**, **Mantine UI**, and **Vercel**, driving user acquisition through targeted campus marketing to onboard **45+** registered users
- Designed a 3NF normalized PostgreSQL schema with Docker deployment on a local Linux server; secured Vercel API communications by implementing **mTLS** with a custom Certificate Authority and DDNS via the **Cloudflare API**
- Developed advanced trip-matching and safety features utilizing **PostGIS** for geospatial querying and **Google Maps/Places APIs** to provide privacy-preserving fuzzy location sharing
- Built live messaging and state tracking with **Firebase Cloud Messaging**, processing **75+** real-time events and ensured data resilience by automating **pg_dump** remote backups via **cron jobs** and **rsync**

Schedule Web App lhsschedule.netlify.app

July 2020 - December 2024

- Developed a successful high school scheduling web app using **ReactJS**, **Material UI**, and **Firebase**, seamlessly handling complex real-time schedule adjustments and features like multi-device syncing, offline-loading, PDF-scanning, and friends
- Supported **1,400+** registered users and **900** daily active users consistently during peak back-to-school traffic by scaling the architecture on **Netlify**

Awards

- 1st Place** (Beginner) & **6th/131 Overall** (Advanced) at UIUC SIGPwny Fall 2025 Capture The Flag Cybersecurity Competition 2025.fallctf.com ([team null](#))
September 21, 2025

Skills

Languages: Python, Java, C++, SQL, HTML/CSS, JavaScript, TypeScript, Kotlin, Bash

Frameworks & Tools: PyTorch, LangChain, React, Next.js, Docker, PostgreSQL, pgvector, PostGIS, ChromaDB, Git, GitLab CI/CD, Azure, Firebase, Linux, Tailscale, Android Studio